Experiment No. 07: Demonstration of Router Hardware and Connecting Different LANs

Aim

To identify the hardware interfaces of a Router and to design, configure, and verify the connectivity between two distinct Local Area Networks (LANs) using a Router in Cisco Packet Tracer.

Theory

What is a Network Router?

Definition: A router is a networking device operating at the Network Layer (Layer 3) that connects multiple, often disparate, networks together.Function: It analyzes data packets (IP addresses) to determine the best route for them to travel to their destination, moving data between LANs and the Internet.Capabilities: Routers allow multiple devices to share a single Internet connection.

The Role of a "Default Gateway"

Definition: The default gateway is the node in a computer network that serves as an access point to another network, usually the router's interface connected to the local network.Role: When a device (computer, phone) wants to send data to an IP address outside its own subnet (e.g., accessing a website), it sends the packets to the default gateway.Function: It is the "exit door" for all traffic destined for remote networks.

How a Router Differs from a Switch

Layer of Operation: Routers operate at Layer 3 (Network Layer) using IP addresses, while switches operate at Layer 2 (Data Link Layer) using MAC addresses.Function: Switches connect devices within the same network (LAN) to share resources. Routers connect different networks together.

Traffic Management: Switches use a MAC address table for communication; routers use a routing table.

What is a Routing Table and How It Works

Definition: A routing table is a data table stored in a router's RAM that lists routes to particular network destinations, including interfaces, next-hop IP addresses, and metrics.How it Forwards Packets:Packet Arrival: A packet arrives at the router, and the router examines the destination IP address.Table Lookup: The router searches its table for the best matching route (longest prefix match) to that destination network.Forwarding: Once the match is found, the router determines the outgoing interface and forwards the packet toward the next hop.Default Route: If no specific route matches, the router uses the "default route" (0.0.0.0/0) to send the packet to the default gateway.

Hardware Components Required

- **PC-PT (x6):** End devices acting as network nodes (3 for LAN 1, 3 for LAN 2).
- **Switch 2960 (x2):** Two 24-port switches to create independent local networks. One switch for each LAN.
- **Router ISR 4321 (x1):** The device to connect the two LANs.
- **Cabling:** Copper Straight-Through cables for all connections.

Hardware and Connection Setup

- LAN 1 (Network ID): 192.168.1.0 | Subnet Mask: 255.255.255.0
- LAN 2 (Network ID): 10.0.0.0 | Subnet Mask: 255.0.0.0
- PC Naming Convention:
  - LAN 1: S11_01_A, S11_01_B, S11_01_C (IPs: .2, .3, .4)
  - LAN 2: S11_01_D, S11_01_E, S11_01_F (IPs: .2, .3, .4)

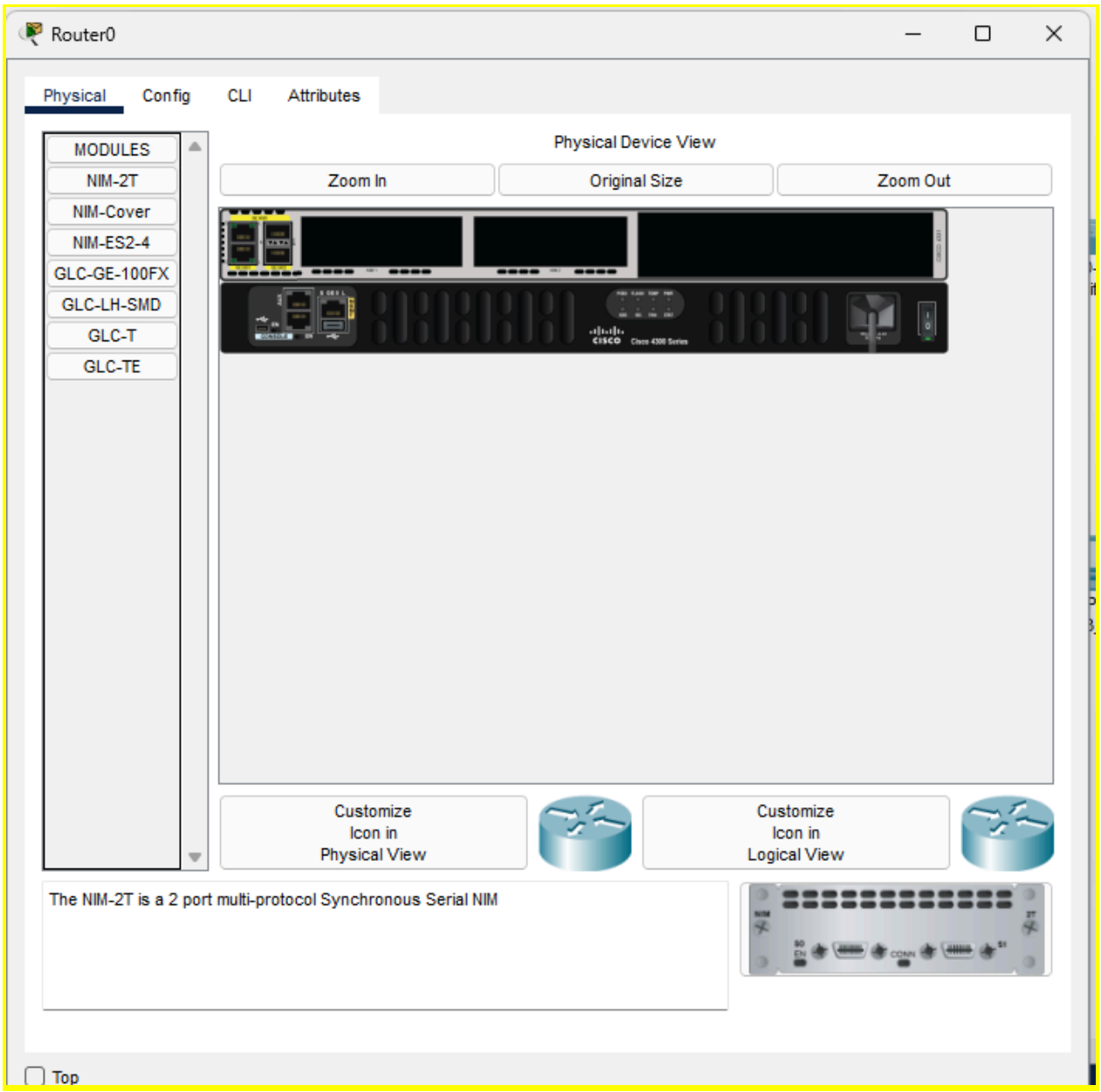
Procedure

Step 1: Physical Topology Design

- Place the ISR 4321 Router in the center of the workspace.
- Place two 2960 Switches (one for each LAN) and six PCs (three per switch).
- Naming: Rename PCs following the convention (e.g., S11_01_A through S11_01_F).

Step 2: Hardware Identification

- Click on the Router and select the Physical tab.
- Observe the modular slots and the GigabitEthernet ports.
- Note the power switch; ensure the router is turned ON.
- (Task: Paste a screenshot of the physical rear view of router



Step 3: Physical Cabling

- Connect Switch 0 to Router interface GigabitEthernet 0/0/0.
- Connect Switch 1 to Router interface GigabitEthernet 0/0/1.
- Connect all PCs to their respective switches.
- Observation: Note that the link lights between the switches and the router are Red. This is because router ports are disabled by default.

Step 4: Router Interface Configuration (The "Bridge" Setup)

1. Click **Router > CLI**. Type ‘no’ at the initial setup prompt.

2. Enter the following commands to configure the gateways:
    - Router> enable
    - Router# config t
    - Router(config)# interface gigabitEthernet 0/0/0
    - Router(config-if)# ip address 192.168.1.1 255.255.255.0
    - Router(config-if)# no shutdown
    - Router(config-if)# interface gigabitEthernet 0/0/1
    - Router(config-if)# ip address 10.0.0.1 255.0.0.0
    - Router(config-if)# no shutdown

Observation: The link lights should now turn Green.

Command	Purpose / Function
enable	Moves the user from **User EXEC Mode** (view only) to **Privileged EXEC Mode**. This allows the user to view detailed configurations and enter setup modes.
config t (configure terminal)	Moves the router into **Global Configuration Mode**. This is the only mode where you can make changes that affect the router as a whole.
interface gigabitEthernet 0/0/0	Moves the router into **Interface Configuration Mode**. It tells the router, "The next commands I type are only for this specific physical port."
ip address [IP] [Mask]	Assigns a logical Identity (IP) and defines the network size (Subnet Mask) for that interface. This becomes the **Default Gateway** for that LAN.
no shutdown	By default, Cisco router ports are "Administratively Down" (turned off for security). This command **powers on** the interface to allow data flow.

Step 5: IP Addressing (Logical Setup)

- **For LAN 1 PCs:** Set IPs 192.168.1.x and set Default Gateway to 192.168.1.1.
- **For LAN 2 PCs:** Set IPs 10.0.0.x and set Default Gateway to 10.0.0.1.

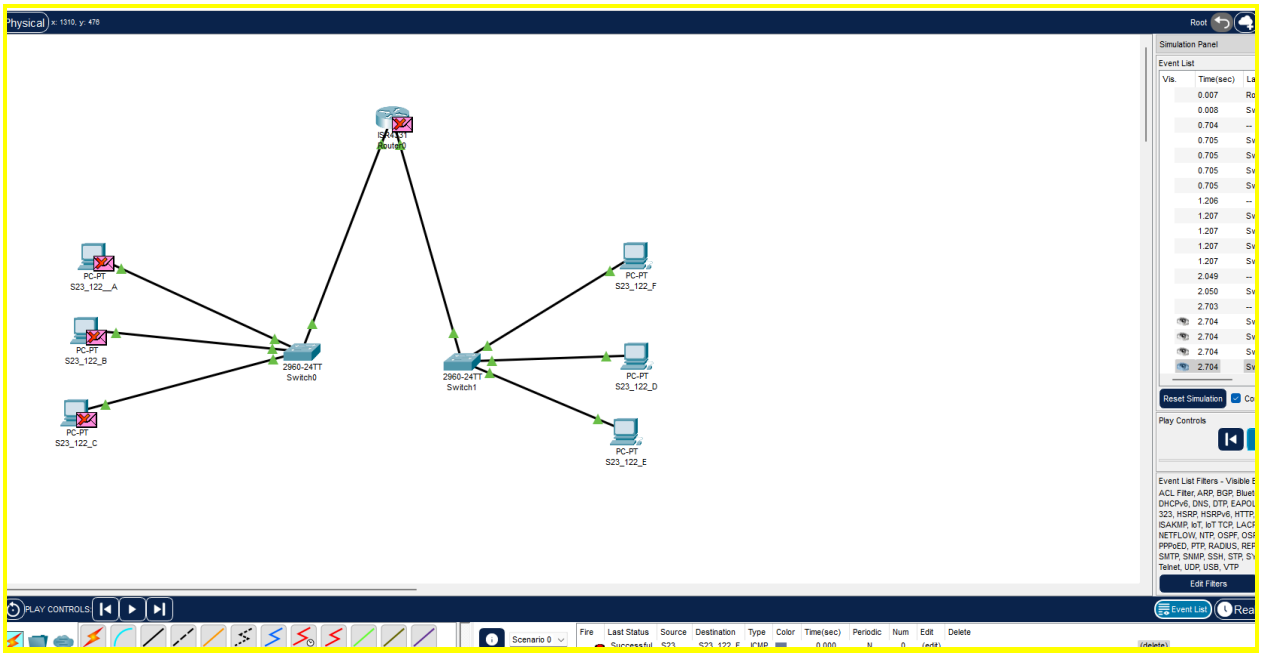
Step 6: Testing and Simulation

1. **Real-time:**
    - Open Command Prompt on **S11_01_A** and type ping 10.0.0.4.

- Verify connectivity. (Note: The first ping may timeout due to ARP).

2. **Simulation Mode:**
   - Filter for ICMP. Use the Add Simple PDU (Envelope) to click S11_01_A (LAN 1) then S11_01_F (LAN 2).
   - Observe the packet crossing from the 192.x network to the 10.x network through the router.

Task: Paste the screenshot (full screen) of the successful simulation showing the packet at the Router.



Observations

- **Gateway Necessity:** If the "Default Gateway" is not configured on the PCs, the packet never reaches the router and communication fails.
- **Routing Decision:** The Router stops the broadcast from LAN 1 and intelligently forwards the packet specifically to the interface connected to LAN 2.
- **Different Subnets:** Communication was successfully established between two completely different IPs, which is impossible with only a switch.

Conclusion

The experiment successfully demonstrates that a **Router** is the essential hardware for Inter-LAN or Inter-Network communication. It proves that while switches manage local traffic, routers are required to cross network boundaries by using logical IP addresses.